PLDM Modeling for Crash Dump, Bulk Telemetry and Watchdog/Timer

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- This information is a summary of the information that will appear in the specifications. See the specifications for further details.

Three Data Models, One PMCI Direction

- The three proposals extend existing PLDM capabilities to standardize how platforms expose important operational data and controls
- Leverages existing PMCI specifications such as DSP0240 PLDM Base Specification, DSP0242 PLDM for File Transfer, and DSP0248 PLDM for Monitor and Control, rather than inventing new transport protocols
- Keep vendor-defined telemetry and crash data content flexible while standardizing discovery, control, and transfer of the content
- Bulk Telemetry, Crash Dump, Timer/Watchdog: Three different operations, single manageability model

Bulk Telemetry Objective

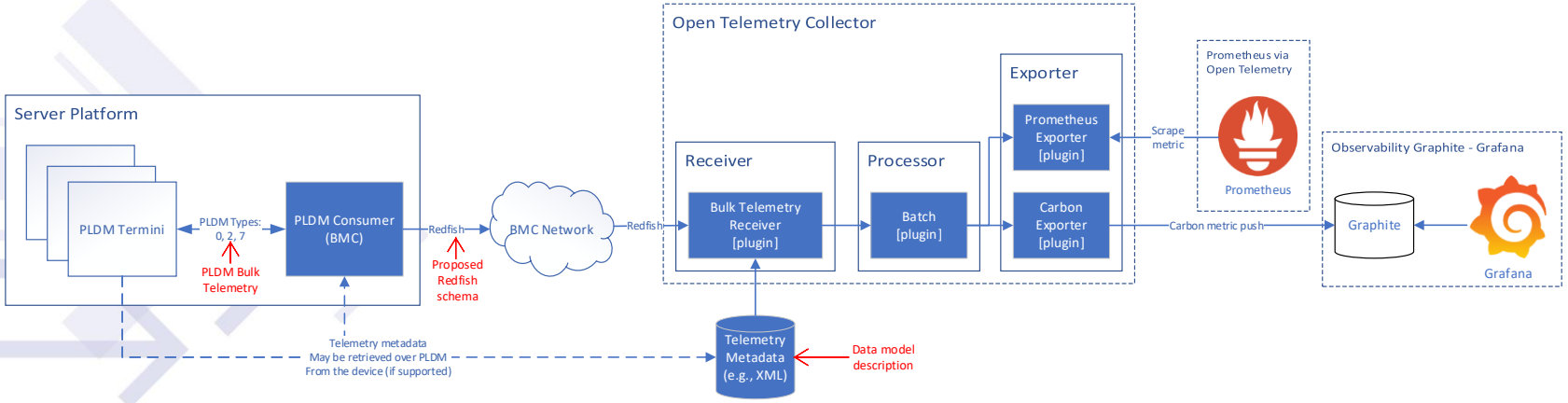
- Standardize the PLDM Data and Operational model for discovery, control, and content (data) transfer
- Metadata and telemetry properties are defined by standard bodies or by the supplier.
- The transfer is standard PLDM for File Transfer
- The association of the describing (Metadata) files to the data files is also standard PLDM for File Transfer Data Model

Bulk telemetry: four data behaviors

- Telemetry file: collection of device data samples, typically processed by an analyzing tool such as Promethius or Graphite
- Snapshot: filtered unit of telemetry captured at a defined sampling period.
- Streaming: Data is transferred as it is produced using a PLDM for File Transfer SerialTxFIFO
- Telemetry Events: captures telemetry events which are typically different than manageability events and may be associated with the streaming telemetry. This is expected to be an independent data stream from the producer
- Optional metadata files are supported when the device can provide them

Example usage

- OpenTelemetry.org use case – data processing at the OT collector
- BMC consumes telemetry using PLDM Bulk Telemetry model
  - BMC does not process the telemetry
  - BMC exposes bulk telemetry over Redfish



Crash Dump (Log) Objective

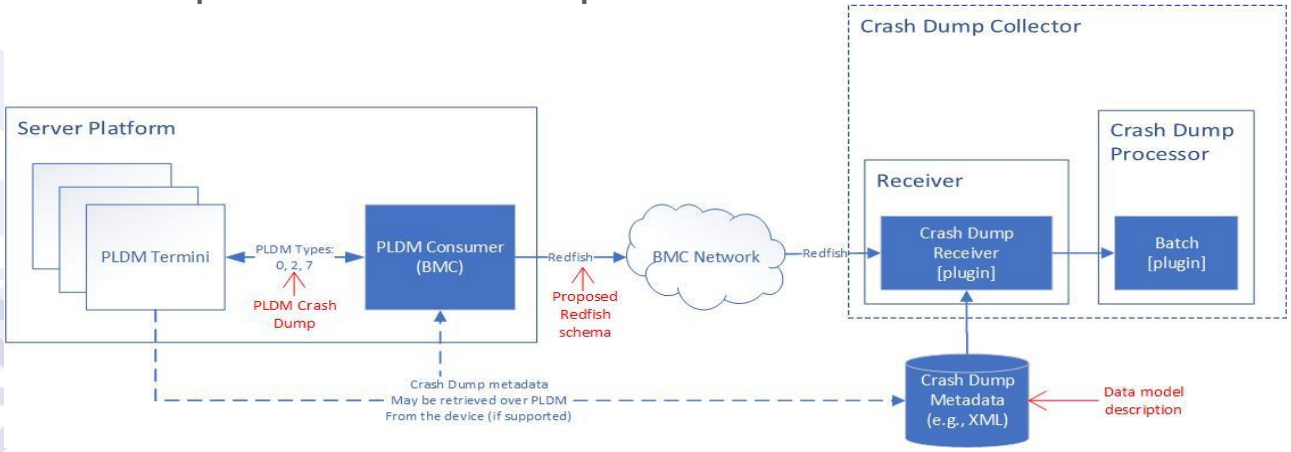
- Define an informative industry method to create, control, and collect crash dump files from an entity either in an error state or manually created
- Standardize the PLDM Data and Operational model for discovery, control, and content (data) transfer
- Metadata and telemetry properties are defined by standard bodies or by the supplier.
- The transfer is standard PLDM for File Transfer
- The association of the describing (Metadata) files to the data files is also standard PLDM for File Transfer Data Model

Typical Crash File Log Data Model

- Vendor (supplier) or other standard body defined format
- Allows the creation to be event (error) or manually created (triggered)
- Allows a single file or associated multiple files (objects)
- Allows a single event instance or repeated (multiple) crash file log instances

Example usage

- OpenTelemetry.org use case – data processing at the OT collector
- BMC collects the Crash Dump using PLDM for File Transfer
  - BMC does not process the Crash Dump
  - BMC exposes Crash Dump over Redfish



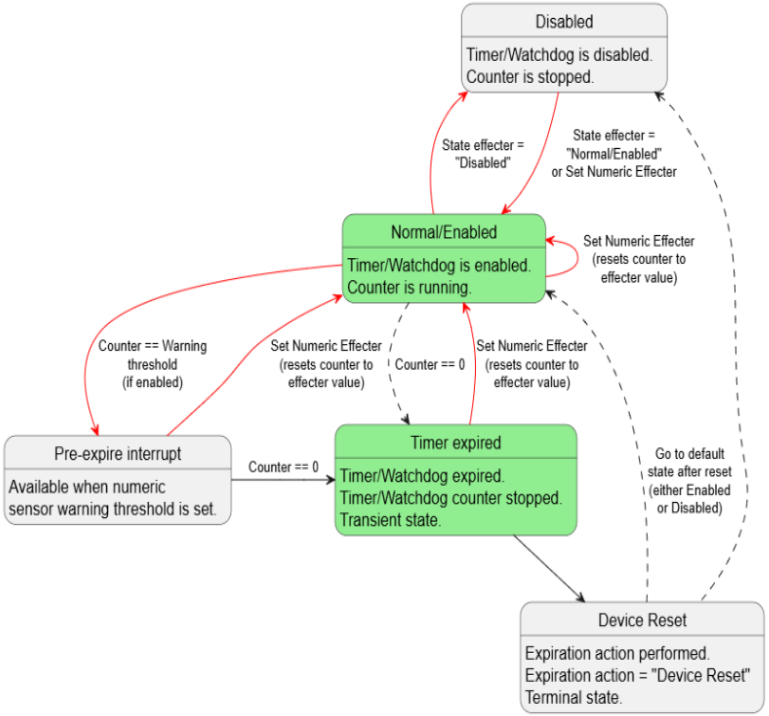
Timer and Watchdog Objective

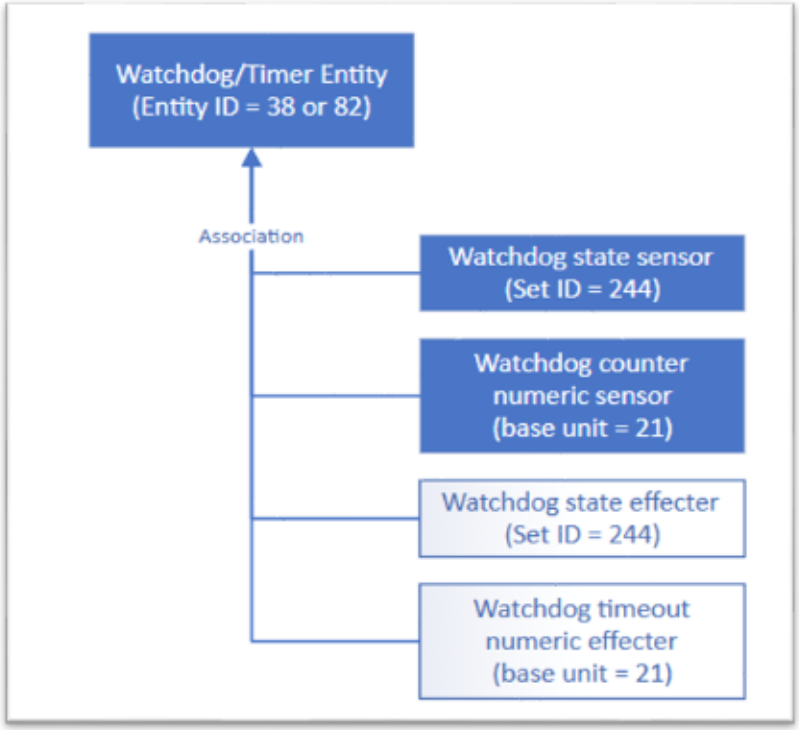
- Describe a flexible PLDM timer model based on DSP0248 PLDM Monitor and Control defined entities, sensors, effecters, and state sets.
- Allows a device to describe timers and watchdogs that may trigger actions upon expiration
- Optionally, a device may permit a PLDM Initialization Agent to control (or set) a timer or watchdog.
- The action taken upon expiration is device vendor defined and may be a PLDM event, a Telemetry event or initiate a Crash Dump

Timer and Watchdog Model Definition

- A Timer is typically a count down but may be a count up clock that optionally may trigger an event or take an action at expiration. A timer is not periodically refreshed.
- A watchdog is a countdown clock periodically refreshed by an agent and takes an action at expiration.
- Both can be modelled using the DSP0248 PLDM for Monitor and Control specification

PLDM DSP0248 Timer Model Example



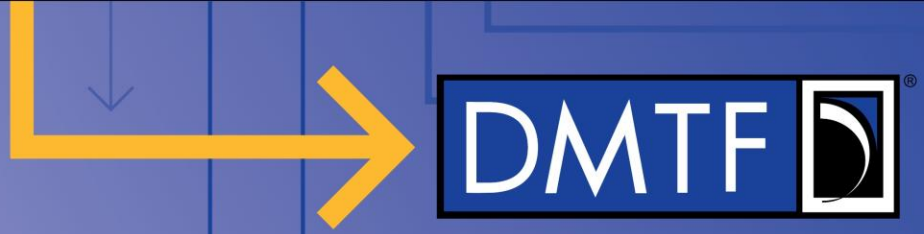


External connections and use cases

- Redfish Alignment:
  - Bulk Telemetry aligns with TelemetryService/TelemetryData
  - Crash Log aligns with LogService
  - Timer and Watchdog aligns with HostWatchdogTimer
- OpenTelemetry receives device telemetry data using out of band BMC collection and transferred to OpenTelemetry service. The device telemetry data is opaque to the BMC
- These are informative alignment points, not replacements for the PMCI model requirements.

Call for Action

- Need explicitly defined Redfish Data Models to move Crash Dump and Bulk Telemetry files from the platform to the entity hosting the analytic tools that operate on the data
  - Enhancements to existing schema to simplify BMC PLDM to Redfish transfers.
- Join a DMTF Work Group and bring enhancements
- Follow the DMTF Standards site:
  - https://www.dmtf.org/standards/pmci
- Use the Feedback Portal to submit requests
  - https://www.dmtf.org/standards/feedback



For more information, visit dmtf.org

Learn about the PMCI working group at dmtf.org/standards/pmci

Thank you!